

Folha 5

Exercício 5.1

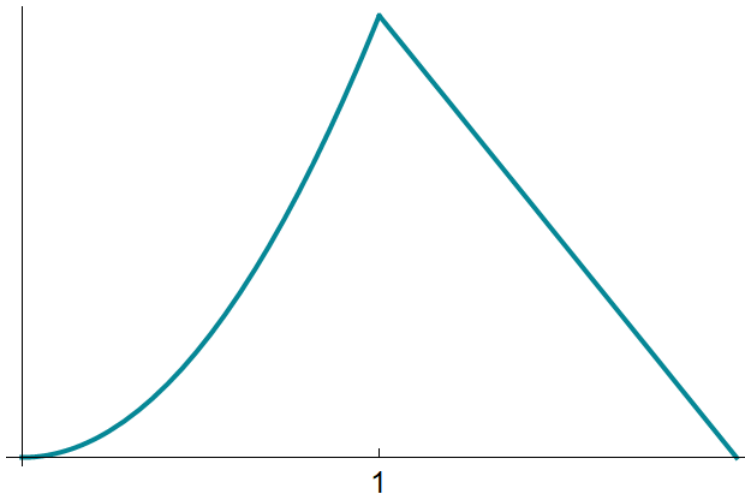
a

-2

b

$y = -1 - 2 x$

Exercício 5.2



Exercício 5.3

a

$D[2 x^3 - x^2 + 7, x]$

$-2 x + 6 x^2$

b

$D[(6 x + 1)^5, x]$

$30 (1 + 6 x)^4$

c

$D[\text{Sqrt}[x] + x^\pi, x]$

$\frac{1}{2 \sqrt{x}} + \pi x^{-1+\pi}$

d

$D[-x / \text{Sqrt}[x], x]$

$-\frac{1}{2 \sqrt{x}}$

e

$D[1 / x^2, x]$

$-\frac{2}{x^3}$

f

$D[x^{(2/3)}, x]$

$\frac{2}{3 x^{1/3}}$

g

$D[x^3 / (x^2 - 4), x] // \text{Simplify}$

$\frac{x^2 (-12 + x^2)}{(-4 + x^2)^2}$

h

i

D[Exp[x] / (x + 1), x] // Simplify

$$\frac{e^x x}{(1 + x)^2}$$

j

D[x^3 Exp@x, x]

$$3 e^x x^2 + e^x x^3$$

k

D[x Log[x^2 + x + 1], x] // Simplify

$$\frac{x (1 + 2 x)}{1 + x + x^2} + \text{Log}\left[1 + x + x^2\right]$$

l

D[Sin@x + Cos@x, x]

$$\text{Cos}\left[x\right] - \text{Sin}\left[x\right]$$

m

D[Tan@x, x]

$$\text{Sec}\left[x\right]^2$$

n

D[5 x^3 Cos[2 x], x]

$$15 x^2 \text{Cos}\left[2 x\right] - 10 x^3 \text{Sin}\left[2 x\right]$$

o

D[Exp@x Sin@x / Log@x, x] // Simplify

$$\frac{e^x (x \text{Cos}\left[x\right] \text{Log}\left[x\right] + (-1 + x \text{Log}\left[x\right]) \text{Sin}\left[x\right])}{x \text{Log}\left[x\right]^2}$$

p

D[Exp[Sin[x]], x]

$$e^{\text{Sin}\left[x\right]} \text{Cos}\left[x\right]$$

q

D[Sin[Cos[x^2]], x]

$$-2 x \text{Cos}\left[\text{Cos}\left[x^2\right]\right] \text{Sin}\left[x^2\right]$$

r

D[x^(-2/3) Exp@x Sin@x, x]

$$\frac{e^x \text{Cos}\left[x\right]}{x^{2/3}} - \frac{2 e^x \text{Sin}\left[x\right]}{3 x^{5/3}} + \frac{e^x \text{Sin}\left[x\right]}{x^{2/3}}$$

Exercício 5.4

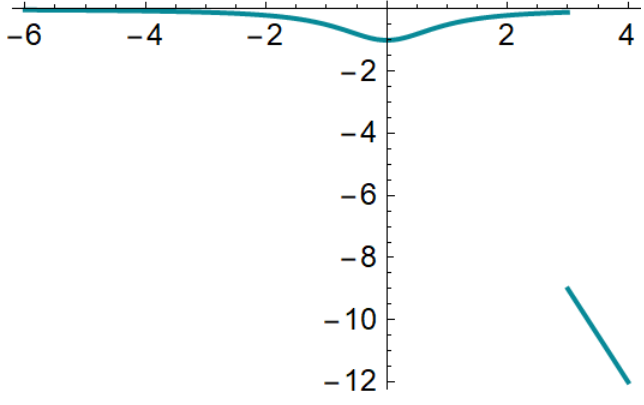
a

$$(\theta, \theta)$$

b

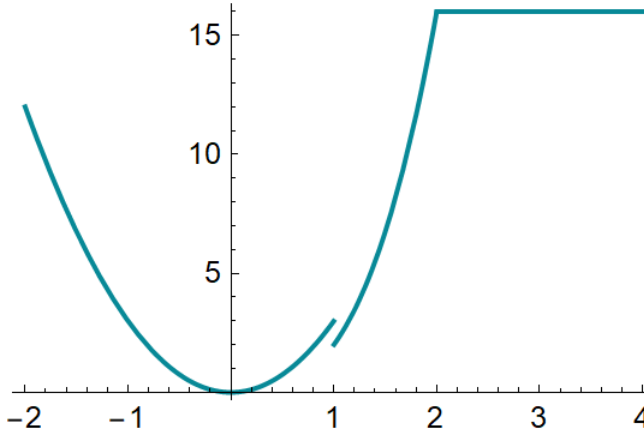
$$y = 1 - e + \frac{-1 + x}{e}$$

$$f[x] = \begin{cases} -\frac{1}{1+x^2} & x < 3 \\ -3x & x \geq 3 \end{cases}$$



$$f' [x] = \begin{cases} \frac{2x}{(-1-x^2)^2} & x < 3 \\ -3 & x > 3 \end{cases}$$

$$g[x] = \begin{cases} 3x^2 & x \leq 1 \\ 2x^3 & 1 < x < 2 \\ 16 & x \geq 2 \end{cases}$$



$$g' [x] = \begin{cases} 6x & x < 1 \\ 6x^2 & 1 < x < 2 \\ 0 & x > 2 \end{cases}$$

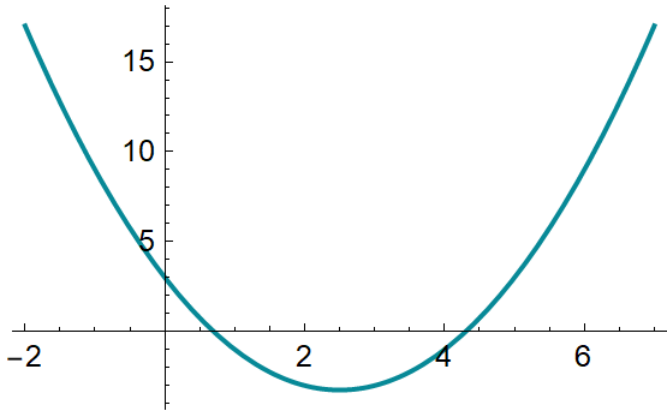
Exercicio 5.6

a = 27, b = -54

Exercicio 5.7

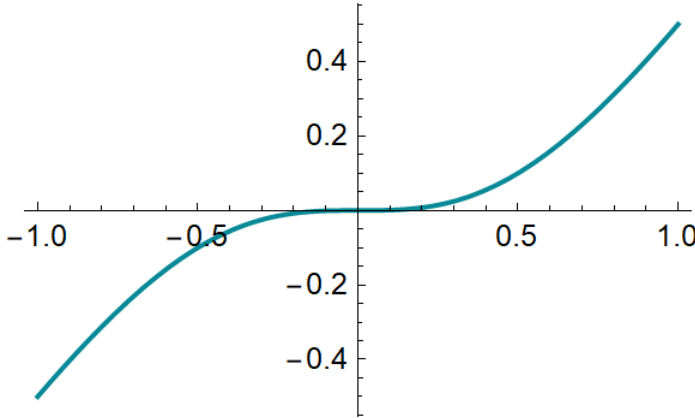
a

`Plot[x^2 - 5 x + 3, {x, -2, 7}]`



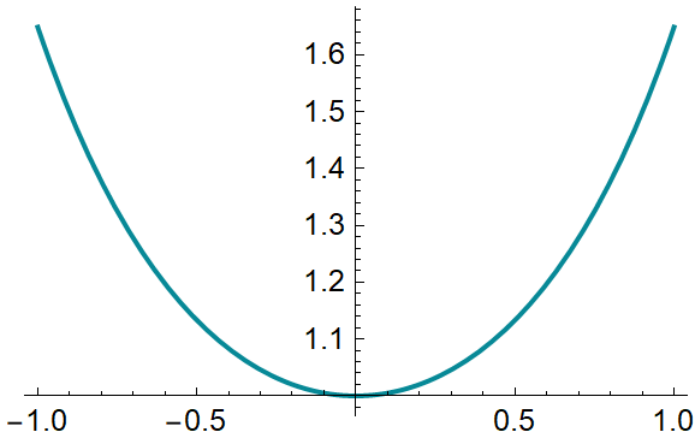
b

`Plot[x^3 / (x^2 + 1), {x, -1, 1}]`



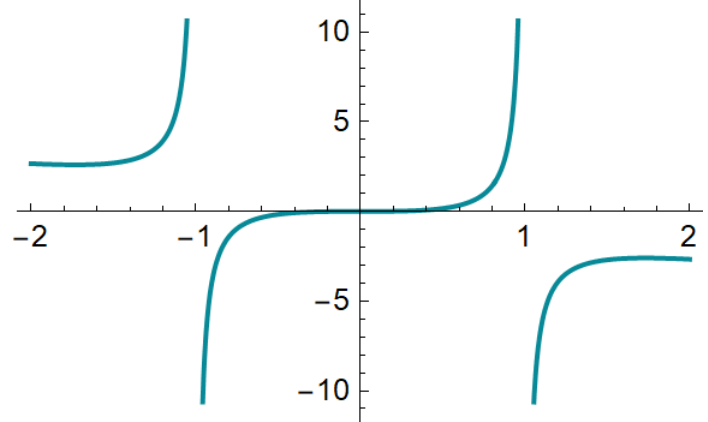
c

`Plot[Exp[x^2 / 2], {x, -1, 1}]`



d

`Plot[x^3 / (1 - x^2), {x, -2, 2}]`



Exercício 5.8

2

Exercício 5.9

$-\frac{3}{2}$

Exercício 5.10

a

$y = x+1$

b

Sim

c

$-\frac{\text{Log}[2]}{8}$

Exercício 5.14

b

$b < -5$

Exercício 5.15

b

$-2 < b < 2$

Exercício 5.16

Não

Exercício 5.21

a

$\text{Limit}[\text{f1}[x], x \rightarrow -\text{Infinity}]$

$\emptyset$

$\text{Limit}[\text{f2}[x], x \rightarrow \text{Infinity}]$

$\frac{\pi}{2}$

c

f é decrescente em $]-\infty, -1]$ e é crescente em $[-1, +\infty[$.

d

$f(\mathbb{R}) = \left[-\frac{1}{e}, \frac{\pi}{2}\right]$

Exercício 5.22

a

$\text{Limit}\left[\frac{4x^2 - \text{Tan}[x]^2}{\text{Cos}[x]^2 - 1}, x \rightarrow 0\right]$

-3

b

Limit[$\frac{1}{\text{Log}@\text{x}} - \frac{1}{\text{x} - 1}, \text{x} \rightarrow 1$]

$\frac{1}{2}$

c

Limit[$\text{x Log}@\text{x}, \text{x} \rightarrow 0$]

0

d

Limit[$\text{x}^{\text{x}}, \text{x} \rightarrow 0$]

1

e

Limit[$\text{Cos}[\text{x}]^{\frac{1}{\text{x}^2}}, \text{x} \rightarrow 0$]

$\frac{1}{\sqrt{e}}$

f

Limit[$\frac{\text{Exp}[\text{x}] + \text{Exp}[-\text{x}] - 2}{1 - \text{Cos}[\text{x}]}, \text{x} \rightarrow 0$]

2

g

Limit[$\frac{\text{x Cos}@\text{x} - \text{Sin}@\text{x}}{\text{x}^3}, \text{x} \rightarrow 0$]

$-\frac{1}{3}$

h

Limit[ArcSin[x] Cot[x], $\text{x} \rightarrow 0$]

1

i

Limit[$\frac{\text{Log}[\text{Sin}[5 \text{x}]]}{\text{Log}[\text{Sin}[6 \text{x}]]}, \text{x} \rightarrow 0$]

1

j

Limit[$\frac{\text{Sin}@\text{x} - \text{x}}{\text{x Sin}@\text{x}}, \text{x} \rightarrow 0$]

0

k

Limit[$\frac{\text{Cos}[3 \text{x}] - 1 + 5 \text{x}^2}{\text{x}^3}, \text{x} \rightarrow 0$]

∞

l

Limit[$\frac{\sqrt{\text{x}} - \text{Sin}[\sqrt{\text{x}}]^2}{\text{x}}, \text{x} \rightarrow 0$]

∞

m

$$\text{Limit} \left[\frac{\sin[x^3] - \sin[x]^3}{x^3}, x \rightarrow 0 \right]$$

④

n

$$\text{Limit} \left[\frac{1}{x} - \frac{1}{\sin x}, x \rightarrow 0 \right]$$

○

$$\text{Limit} \left[\frac{\sinh x - \sin x}{x^3}, x \rightarrow 0 \right]$$

$\frac{1}{3}$

p

$$\text{Limit} \left[\frac{x - \sin x}{x^3}, x \rightarrow 0 \right]$$

$\frac{1}{6}$

q

$$\text{Limit} \left[\frac{\sin[4x] \sin[3x]}{x \sin[2x]}, x \rightarrow \pi \right]$$

④

r

$$\text{Limit} \left[\frac{3x}{x + 2 \sin[x]}, x \rightarrow \infty \right]$$

3

Exercício 5.24

- a** existe
b não existe
c existe
d não existe
e existe

Exercício 5.25

- | | |
|----------|---|
| a | F |
| b | F |
| c | V |
| d | V |